

Activity 0.1.4 Construct, Test, and Evaluate

PLTW

CONSTRUCT, TEST, AND EVALUATE



Prototyping



Build



Test



Evaluate Solution



Conclusion

ACTIVITY SUPPORTS



Sources

Prototyping

GOALS	MATERIALS	RESOURCES
- Fabricate a prototype using the materials available to you. - Explore design alternatives through the creation and testing of prototypes. - Identify flaws and potential enhancements of a proposed design solution.		
GOALS	MATERIALS	RESOURCES
- Paper - Cardstock - Foam board - Cardboard - Tape		

- Elastic string
- Scissors
- String or yarn

GOALS	MATERIALS	RESOURCES
• <u>Activity 0.1.4 Construct, Test, and Evaluate (Downloadable PDF)</u>		

Now that you and your teammates have selected a solution, you can build a **prototype** that brings your design to life. Engineers and designers use prototypes to understand the working aspects of a solution before manufacturing. After they are built, prototypes are tested to determine feasibility. If issues arise during testing, engineers repeat the design process until they find an optimal solution. Example prototypes are shown in Figures 1 and 2.



Figure 1. Nintendo GameCube Controller Prototype¹

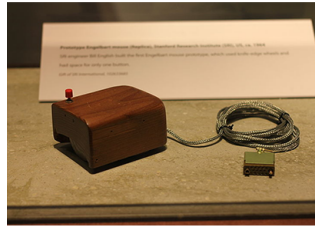


Figure 2. Douglas Engelbart's Prototype Mouse²

Reflection



Is a prototype the same thing as a model? Pause to formulate your opinion before checking your response.

Compare Your Thoughts

People often use the words prototype and model interchangeably, but they are not the same. A model is used to demonstrate or explain how a product will look or feel, while a prototype is used to test working aspects of a product before design finalization.

Build

You are ready to build your first prototype as a team.

- 1 Review the solution you wrote in your engineering notebook.
- 2 Open the concept sketches you created in Activity 0.1.3 Choosing the Best Solution. Use your sketches as a point of reference during today's work.
- 3 Gather the materials you need.
- 4 Begin building your first prototype.

Here the prototype construction process stops briefly so you can get feedback from another team in your class.

- 5 Show your prototype to another team. Explain its purpose and describe the challenges you face. Work together to brainstorm possible solutions to the problems.

NOTE: Listen carefully. Document the suggestions your classmates give you.

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Resume building your prototype.

Test

After you create your prototype, you need to test its performance, usability, or durability. During testing, you will likely experience failure, which is a great thing! A philosophy often repeated and credited to Tom Kelley of IDEO, a successful worldwide engineering design and innovation consulting firm, is “Fail often to succeed sooner.” Through failure, you will gain knowledge, tenacity, and the motivation to try again.

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With your team, create one test to evaluate a part of your design. Be sure to identify the independent and dependent variables in the test.

8

In your engineering notebook, thoroughly describe the testing procedures. The team you worked with during the feedback-gathering exercise will be conducting the tests on your prototype. Leave the testing procedures out for their review.

9

Swap your prototype with the testing team.

10

Perform the given tests on your classmates' prototype.

11

Provide the following feedback below the team's testing procedures in the notebook.

- Is the prototype functional? What works? What does not work?
- What improvements could you make?

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Retrieve your prototype. Study the feedback to determine whether your prototype meets the criteria and constraints defined in your design statement.

Evaluate Solution

As soon as you find out (through research, testing, or expert knowledge) that your current design will not meet your design statement's criteria, return to a previous step in the Engineering Design Process (Figure 3) to rethink and refine the design solution to better address the design requirements.

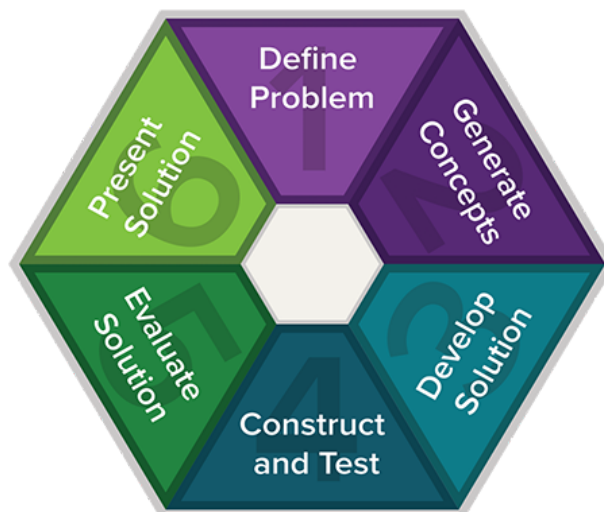


Figure 3. Engineering Design Process

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Improve your design based on the feedback you were given.

Turn and Talk



Was the feedback you received helpful? Were you surprised by any of the testing results?

CAREER CONNECTION

Michelle Brus | *Test Engineer*

Michelle Brus became interested in engineering during an 8th-grade career research project. In high school, she took five PLTW classes: Introduction to Engineering Design, Principles of Engineering, Civil Engineering and Architecture, Computer Integrated Manufacturing, and Engineering Design and Development. Her high school experiences led her to major in Mechanical Engineering at Iowa State.

During her junior year of college, Michelle took a semester off to intern at Whirlpool as a Supplier Quality Engineer for seven months. She gained vast experience in engineering and manufacturing but credits the internship most for teaching her early on that she did not enjoy quality engineering enough to make it a lifelong career.



Michelle now works as a Test Engineer for John Deere. Her job has allowed her to work hands-on with the machines as she validates the durability and performance of the construction machines' instrumentation. This includes parts like seats, joysticks, steering wheels, doors, and windows. Michelle creates the test plans, performs the tests, documents

the results, and works with the design team to propose improvements when needed.

According to Michelle, people skills are the most important thing to develop as a young adult. Technical skills can be learned on the job, but any employer will want you to already have a strong willingness to learn and the ability to work well with others.



Conclusion

Question 1

How did your prototype evolve throughout the iteration process?

Question 2

What did you learn by testing another group's prototype?

Sources

Third-party images, videos, and Historical Highlight information used throughout this activity were drawn from the following sources:

1. Nintendo GameCube Dolphin DS5 Controller Prototype, Evans-Amos, Wikimedia, permitted under Public Domain.
2. Douglas Engelbart's prototype mouse - Computer History Museum, Michael Hicks, Wikimedia, permitted under [**CC-BY 2.0**](#).